

# Modified Delta Results

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# Quantization Error Accumulation in Old Delta Formula

$$\Delta X_t^L = X_t^L - X_{t-1}^L$$

$$\Delta X_{q_t}^L = \Delta X_t^L + q_{\text{err}_t}$$

$$Y_t^L = Y_{t-1}^L + W^L \Delta X_{q_t}^L$$

$$\mathbf{X}_t^L := \mathbf{X}_t^L$$

$$\begin{aligned} Y_t^L &= Y_{t-1}^L + W^L (X_t^L - X_{t-1}^L) + W^L q_{\text{err}_t} \\ &= \left[ Y_{t-2}^L + W^L (X_{t-1}^L - X_{t-2}^L) + W^L q_{\text{err}_{t-1}} \right] + W^L (X_t^L - X_{t-1}^L) + W^L q_{\text{err}_t} \\ &= Y_{t-2}^L + W^L (X_t^L - X_{t-2}^L) + W^L q_{\text{err}_{t-1}} + W^L q_{\text{err}_t} \end{aligned}$$

$$Y_t^L = Y_0^L + W^L (X_t^L - X_0^L) + \sum_{i=1}^t W^L q_{\text{err}_i}$$

$$= W^L X_t^L + \sum_{i=1}^t W^L q_{\text{err}_i}$$

# New Delta Formula

$$\Delta X_t^L = X_t^L - \widehat{X}_{t-1}^L$$

$$\Delta X_{q_t}^L = \Delta X_t^L + q_{\text{err}_t}$$

$$Y_t^L = Y_{t-1}^L + W^L \Delta X_{q_t}^L$$

$$\widehat{\mathbf{X}}_t^L := \widehat{\mathbf{X}}_{t-1}^L + \Delta \mathbf{X}_{q_t}^L$$

$$\begin{aligned} Y_t^L &= Y_{t-1}^L + W^L \left( X_t^L - \widehat{X}_{t-1}^L \right) + W^L q_{\text{err}_t} \\ &= Y_{t-1}^L + W^L \left( X_t^L - X_{t-1}^L - q_{\text{err}_{t-1}} \right) + W^L q_{\text{err}_t} \\ &= \left[ W^L X_{t-1}^L + W^L q_{\text{err}_{t-1}} \right] + W^L \left( X_t^L - X_{t-1}^L \right) - W^L q_{\text{err}_{t-1}} + W^L q_{\text{err}_t} \end{aligned}$$

$$\begin{aligned} Y_t^L &= W^L X_{t-1}^L - W^L X_{t-1}^L + W^L X_t^L + W^L q_{\text{err}_{t-1}} - W^L q_{\text{err}_{t-1}} + W^L q_{\text{err}_t} \\ &= W^L X_t^L + W^L q_{\text{err}_t} \end{aligned}$$

# DeltaDuQuant and DeltaQuaRot Single-anchor Results

LLaDA-8B-Instruct accuracy: QuaRot vs. DuQuant rotation quantization, with and without delta-quant, at W4A4.  $\Delta\%$  is the relative change of the group average w.r.t. the FP16 base.

| Model             | Setting | Method           | Math  |         |       |            | Code      |       |       |            |
|-------------------|---------|------------------|-------|---------|-------|------------|-----------|-------|-------|------------|
|                   |         |                  | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16    | Base             | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          |
|                   | W4A4    | QuaRot           | 67.55 | 25.34   | 46.45 | ↓18.32%    | 26.83     | 32.00 | 29.41 | ↓25.93%    |
|                   |         | Old DeltaQuaRot  | 49.36 | 16.77   | 33.06 | ↓41.86%    | 21.34     | 31.00 | 26.17 | ↓34.11%    |
|                   |         | DeltaQuaRot      | 71.11 | 28.58   | 49.85 | ↓12.33%    | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         | DuQuant          | 75.74 | 32.75   | 54.25 | ↓4.60%     | 35.37     | 29.80 | 32.58 | ↓17.96%    |
|                   |         | Old DeltaDuQuant | 45.49 | 15.67   | 30.58 | ↓46.22%    | 19.51     | 23.60 | 21.56 | ↓45.73%    |
|                   |         | DeltaDuQuant     | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |

# Uniform Reanchoring Results

LLaDA-8B-Instruct accuracy under delta-quant on top of DuQuant/QuaRot rotations, swept over the number of full-precision anchors spread uniformly across the generation trajectory.

| Model             | Setting | Method       | Anchors | Math  |         |       |            | Code      |       |       |            |
|-------------------|---------|--------------|---------|-------|---------|-------|------------|-----------|-------|-------|------------|
|                   |         |              |         | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16    | Base         | –       | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          |
|                   |         |              | 1       | 71.11 | 28.58   | 49.85 | ↓12.33%    | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         |              | 2       | 70.74 | 28.41   | 49.57 | ↓12.82%    | 31.71     | 36.60 | 34.15 | ↓14.01%    |
|                   |         |              | 4       | 71.27 | 28.99   | 50.13 | ↓11.84%    | 31.71     | 36.00 | 33.85 | ↓14.77%    |
|                   |         |              | 8       | 70.81 | 29.19   | 50.00 | ↓12.06%    | 32.32     | 36.00 | 34.16 | ↓14.00%    |
|                   |         |              | 16      | 71.72 | 29.16   | 50.44 | ↓11.29%    | 35.98     | 36.60 | 36.29 | ↓8.64%     |
|                   |         |              | 32      | 71.49 | 29.51   | 50.50 | ↓11.18%    | 37.80     | 34.80 | 36.30 | ↓8.60%     |
|                   | W4A4    | DeltaQuaRot  | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | 76.57 | 33.36   | 54.97 | ↓3.33%     | 39.63     | 32.60 | 36.12 | ↓9.07%     |
|                   |         |              | 4       | 75.74 | 32.84   | 54.29 | ↓4.52%     | 39.63     | 32.60 | 36.12 | ↓9.07%     |
|                   |         |              | 8       | 76.27 | 33.59   | 54.93 | ↓3.39%     | 37.80     | 32.40 | 35.10 | ↓11.63%    |
|                   |         |              | 16      | 76.42 | 33.62   | 55.02 | ↓3.23%     | 37.80     | 32.80 | 35.30 | ↓11.12%    |
|                   |         |              | 32      | 75.66 | 33.74   | 54.70 | ↓3.80%     | 38.41     | 33.20 | 35.81 | ↓9.85%     |
|                   |         | DeltaDuQuant | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | 76.57 | 33.36   | 54.97 | ↓3.33%     | 39.63     | 32.60 | 36.12 | ↓9.07%     |
|                   |         |              | 4       | 75.74 | 32.84   | 54.29 | ↓4.52%     | 39.63     | 32.60 | 36.12 | ↓9.07%     |
|                   |         |              | 8       | 76.27 | 33.59   | 54.93 | ↓3.39%     | 37.80     | 32.40 | 35.10 | ↓11.63%    |
|                   |         |              | 16      | 76.42 | 33.62   | 55.02 | ↓3.23%     | 37.80     | 32.80 | 35.30 | ↓11.12%    |
|                   |         |              | 32      | 75.66 | 33.74   | 54.70 | ↓3.80%     | 38.41     | 33.20 | 35.81 | ↓9.85%     |

# Exponential Schedule

- For 8 anchors the anchors are placed in timesteps 1,2,4,8,16,32,64, and 128

LLaDA-8B-Instruct accuracy using an exponential (base-2) schedule (anchor positions 1, 2, 4, 8, ..., front-loaded towards the start of the generation trajectory).

| Model             | Setting | Method      | Anchors | Math  |         |       |                      | Code      |       |       |                      |
|-------------------|---------|-------------|---------|-------|---------|-------|----------------------|-----------|-------|-------|----------------------|
|                   |         |             |         | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval | MBPP  | Avg   | $\Delta\%$           |
| LLaDA-8B-Instruct | FP16    | Base        | –       | 78.85 | 34.87   | 56.86 | –                    | 39.63     | 39.80 | 39.72 | –                    |
|                   |         |             | 1       | 71.11 | 28.58   | 49.85 | $\downarrow 12.33\%$ | 31.10     | 36.40 | 33.75 | $\downarrow 15.03\%$ |
|                   |         |             | 2       | 71.19 | –       | –     | –                    | 30.49     | 35.20 | 32.84 | $\downarrow 17.31\%$ |
|                   |         |             | 4       | 71.19 | –       | –     | –                    | 31.71     | 36.00 | 33.85 | $\downarrow 14.77\%$ |
|                   | W4A4    | DeltaQuaRot | 8       | 71.49 | –       | –     | –                    | 34.76     | 35.40 | 35.08 | $\downarrow 11.69\%$ |
|                   |         |             | 1       | 75.82 | 34.09   | 54.95 | $\downarrow 3.36\%$  | 39.02     | 32.60 | 35.81 | $\downarrow 9.84\%$  |
|                   |         |             | 2       | 77.10 | –       | –     | –                    | 37.80     | 32.00 | 34.90 | $\downarrow 12.13\%$ |
|                   |         |             | 4       | 75.59 | –       | –     | –                    | 36.59     | 32.40 | 34.49 | $\downarrow 13.16\%$ |
|                   |         |             | 8       | 76.27 | –       | –     | –                    | 40.85     | 31.40 | 36.13 | $\downarrow 9.05\%$  |

# Offline Calibrated Anchor Scheduling using WikiText2

Table 5: LLaDA-8B-Instruct accuracy under offline-calibrated anchor scheduling.

| Model             | Setting | Method       | Anchors | Math  |         |       |            | Code      |       |       |            |
|-------------------|---------|--------------|---------|-------|---------|-------|------------|-----------|-------|-------|------------|
|                   |         |              |         | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16    | Base         | –       | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          |
|                   |         |              | 1       | 71.11 | 28.58   | 49.85 | ↓12.33%    | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         | DeltaQuaRot  | 2       | 70.66 | 28.67   | 49.67 | ↓12.66%    | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         |              | 4       | 70.20 | 28.87   | 49.54 | ↓12.88%    | 34.15     | 36.00 | 35.07 | ↓11.70%    |
|                   |         |              | 8       | 71.11 | 28.44   | 49.78 | ↓12.46%    | 35.98     | 36.40 | 36.19 | ↓8.89%     |
|                   |         |              | 16      | 71.80 | 28.87   | 50.34 | ↓11.48%    | 38.41     | 35.40 | 36.91 | ↓7.08%     |
|                   |         |              | 32      | 71.11 | 29.11   | 50.11 | ↓11.87%    | 35.37     | 35.00 | 35.18 | ↓11.42%    |
|                   | W4A4    | DeltaDuQuant | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | 75.74 | 33.74   | 54.74 | ↓3.73%     | 42.07     | 32.60 | 37.34 | ↓6.00%     |
|                   |         |              | 4       | 76.95 | 33.59   | 55.27 | ↓2.79%     | 41.46     | 33.00 | 37.23 | ↓6.26%     |
|                   |         |              | 8       | 76.35 | 33.65   | 55.00 | ↓3.27%     | 39.02     | 32.40 | 35.71 | ↓10.09%    |
|                   |         |              | 16      | 75.44 | 33.39   | 54.41 | ↓4.30%     | 41.46     | 32.80 | 37.13 | ↓6.52%     |
|                   |         |              | 32      | 75.59 | 33.33   | 54.46 | ↓4.22%     | 37.20     | 32.40 | 35.20 | ↓11.39%    |

# Offline Calibration On Rotated Activations using WikiText2

Table 12: LLaDA-8B-Instruct accuracy under RQE rotated-eval WikiText2-calibrated anchor scheduling.

| Model             | Setting      | Method      | Anchors | Math  |         |       |                      | Code                |       |                      |                      |
|-------------------|--------------|-------------|---------|-------|---------|-------|----------------------|---------------------|-------|----------------------|----------------------|
|                   |              |             |         | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval           | MBPP  | Avg                  | $\Delta\%$           |
| LLaDA-8B-Instruct | FP16         | Base        | –       | 78.85 | 34.87   | 56.86 | –                    | 39.63               | 39.80 | 39.72                | –                    |
|                   |              |             | 1       | 71.11 | 28.58   | 49.85 | $\downarrow 12.33\%$ | 31.10               | 36.40 | 33.75                | $\downarrow 15.03\%$ |
|                   |              | DeltaQuaRot | 2       | –     | –       | –     | –                    | 31.10               | 36.20 | 33.65                | $\downarrow 15.29\%$ |
|                   |              |             | 4       | –     | –       | –     | –                    | 32.93               | 35.80 | 34.36                | $\downarrow 13.49\%$ |
|                   |              |             | 8       | –     | –       | –     | –                    | 32.32               | 35.00 | 33.66                | $\downarrow 15.26\%$ |
|                   |              |             | 16      | –     | –       | –     | –                    | 34.76               | 35.00 | 34.88                | $\downarrow 12.19\%$ |
|                   |              |             | 32      | –     | –       | –     | –                    | 34.15               | 35.80 | 34.97                | $\downarrow 11.95\%$ |
|                   |              |             | W4A4    | 1     | 75.82   | 34.09 | 54.95                | $\downarrow 3.36\%$ | 39.02 | 32.60                | 35.81                |
|                   | 2            | –           |         | –     | –       | –     | 38.41                | 32.60               | 35.51 | $\downarrow 10.61\%$ |                      |
|                   | DeltaDuQuant | 4           |         | –     | –       | –     | –                    | 37.80               | 33.00 | 35.40                | $\downarrow 10.87\%$ |
|                   |              | 8           |         | –     | –       | –     | –                    | 38.41               | 32.40 | 35.41                | $\downarrow 10.86\%$ |
|                   |              | 16          |         | –     | –       | –     | –                    | 40.85               | 32.60 | 36.73                | $\downarrow 7.54\%$  |
|                   |              | 32          |         | –     | –       | –     | –                    | 38.41               | 33.20 | 35.81                | $\downarrow 9.85\%$  |



# Cross-Task Offline Calibration Results

Table 9: LLaDA-8B-Instruct accuracy under cross-task calibrated anchor scheduling.

| Model             | Setting | Method      | Anchors | Math  |         |       |            | Code      |       |       |            |
|-------------------|---------|-------------|---------|-------|---------|-------|------------|-----------|-------|-------|------------|
|                   |         |             |         | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16    | Base        | –       | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          |
|                   |         |             | 1       | 71.11 | 28.58   | 49.85 | ↓ 12.33%   | 31.10     | 36.40 | 33.75 | ↓ 15.03%   |
|                   |         |             | 2       | 70.74 | 28.88   | 49.81 | ↓ 12.40%   | 31.10     | 36.60 | 33.85 | ↓ 14.78%   |
|                   |         |             | 4       | 71.42 | 29.24   | 50.33 | ↓ 11.49%   | 31.71     | 36.00 | 33.85 | ↓ 14.77%   |
|                   |         |             | 8       | 71.11 | 28.84   | 49.98 | ↓ 12.10%   | 32.93     | 36.20 | 34.56 | ↓ 12.98%   |
|                   |         |             | 16      | 70.13 | 28.68   | 49.40 | ↓ 13.11%   | 32.93     | 35.60 | 34.26 | ↓ 13.74%   |
|                   |         |             | 32      | 68.92 | 29.14   | 49.03 | ↓ 13.77%   | 35.37     | 34.80 | 35.08 | ↓ 11.67%   |
|                   | W4A4    | DeltaQuaRot | 1       | 75.82 | 34.09   | 54.95 | ↓ 3.36%    | 39.02     | 32.60 | 35.81 | ↓ 9.84%    |
|                   |         |             | 2       | 75.89 | –       | –     | –          | 39.63     | 32.60 | 36.12 | ↓ 9.07%    |
|                   |         |             | 4       | 75.44 | –       | –     | –          | 38.41     | 32.80 | 35.61 | ↓ 10.35%   |
|                   |         |             | 8       | 75.97 | –       | –     | –          | 39.02     | 33.00 | 36.01 | ↓ 9.33%    |
|                   |         |             | 16      | 75.13 | –       | –     | –          | 39.63     | 31.80 | 35.72 | ↓ 10.08%   |
|                   |         |             | 32      | 74.91 | –       | –     | –          | 37.20     | 32.40 | 34.80 | ↓ 12.39%   |

# Cross-Task Results on Rotated Activations

Table 13: LLaDA-8B-Instruct accuracy under RQE rotated-eval cross-task calibrated anchor scheduling.

| Model             | Setting | Method       | Anchors | Math  |         |       |            | Code      |       |       |            |
|-------------------|---------|--------------|---------|-------|---------|-------|------------|-----------|-------|-------|------------|
|                   |         |              |         | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16    | Base         | –       | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          |
|                   |         |              | 1       | 71.11 | 28.58   | 49.85 | ↓12.33%    | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         |              | 2       | –     | –       | –     | –          | 31.10     | 36.40 | 33.75 | ↓15.03%    |
|                   |         |              | 4       | –     | –       | –     | –          | 32.32     | 35.80 | 34.06 | ↓14.25%    |
|                   |         |              | 8       | –     | –       | –     | –          | 33.54     | 36.40 | 34.97 | ↓11.96%    |
|                   |         |              | 16      | –     | –       | –     | –          | 33.54     | 36.00 | 34.77 | ↓12.46%    |
|                   |         |              | 32      | –     | –       | –     | –          | 32.93     | 35.80 | 34.36 | ↓13.48%    |
|                   | W4A4    | DeltaQuaRot  | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | –     | –       | –     | –          | 39.63     | 32.60 | 36.12 | ↓9.08%     |
|                   |         |              | 4       | –     | –       | –     | –          | 37.80     | 33.20 | 35.50 | ↓10.62%    |
|                   |         |              | 8       | –     | –       | –     | –          | 37.80     | 33.60 | 35.70 | ↓10.12%    |
|                   |         |              | 16      | –     | –       | –     | –          | 38.41     | 32.60 | 35.50 | ↓10.61%    |
|                   |         |              | 32      | –     | –       | –     | –          | 35.37     | 31.40 | 33.38 | ↓15.95%    |
|                   |         | DeltaDuQuant | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | –     | –       | –     | –          | 39.63     | 32.60 | 36.12 | ↓9.08%     |
|                   |         |              | 4       | –     | –       | –     | –          | 37.80     | 33.20 | 35.50 | ↓10.62%    |
|                   |         |              | 8       | –     | –       | –     | –          | 37.80     | 33.60 | 35.70 | ↓10.12%    |
|                   |         |              | 16      | –     | –       | –     | –          | 38.41     | 32.60 | 35.50 | ↓10.61%    |
|                   |         |              | 32      | –     | –       | –     | –          | 35.37     | 31.40 | 33.38 | ↓15.95%    |
|                   |         |              | 1       | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     |
|                   |         |              | 2       | –     | –       | –     | –          | 39.63     | 32.60 | 36.12 | ↓9.08%     |
|                   |         |              | 4       | –     | –       | –     | –          | 37.80     | 33.20 | 35.50 | ↓10.62%    |
|                   |         |              | 8       | –     | –       | –     | –          | 37.80     | 33.60 | 35.70 | ↓10.12%    |
|                   |         |              | 16      | –     | –       | –     | –          | 38.41     | 32.60 | 35.50 | ↓10.61%    |
|                   |         |              | 32      | –     | –       | –     | –          | 35.37     | 31.40 | 33.38 | ↓15.95%    |

# Online Calibration Results

Table 7: LLaDA-8B-Instruct accuracy under online-calibrated greedy anchor scheduling.

| Model             | Setting | Method       | Anchors | Math  |         |       |                      | Code      |       |       |                      |
|-------------------|---------|--------------|---------|-------|---------|-------|----------------------|-----------|-------|-------|----------------------|
|                   |         |              |         | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval | MBPP  | Avg   | $\Delta\%$           |
| LLaDA-8B-Instruct | FP16    | Base         | –       | 78.85 | 34.87   | 56.86 | –                    | 39.63     | 39.80 | 39.72 | –                    |
|                   |         | DeltaQuaRot  | 1       | 71.11 | 28.58   | 49.85 | $\downarrow 12.33\%$ | 31.10     | 36.40 | 33.75 | $\downarrow 15.03\%$ |
|                   |         |              | 2       | 70.28 | 29.08   | 49.68 | $\downarrow 12.63\%$ | 35.98     | 37.00 | 36.49 | $\downarrow 8.14\%$  |
|                   |         |              | 4       | 70.89 | 28.79   | 49.84 | $\downarrow 12.35\%$ | 34.76     | 37.40 | 36.08 | $\downarrow 9.17\%$  |
|                   |         |              | 8       | 69.98 | 29.89   | 49.93 | $\downarrow 12.19\%$ | 37.20     | 37.20 | 37.20 | $\downarrow 6.35\%$  |
|                   |         |              | 16      | 70.51 | 29.45   | 49.98 | $\downarrow 12.10\%$ | 37.20     | 37.20 | 37.20 | $\downarrow 6.35\%$  |
|                   |         |              | 32      | 70.51 | 29.48   | 49.99 | $\downarrow 12.08\%$ | 37.20     | 37.20 | 37.20 | $\downarrow 6.35\%$  |
|                   | W4A4    | DeltaDuQuant | 1       | 75.82 | 34.09   | 54.95 | $\downarrow 3.36\%$  | 39.02     | 32.60 | 35.81 | $\downarrow 9.84\%$  |
|                   |         |              | 2       | 77.10 | 33.04   | 55.07 | $\downarrow 3.14\%$  | 37.20     | 32.80 | 35.00 | $\downarrow 11.89\%$ |
|                   |         |              | 4       | 76.72 | 32.78   | 54.75 | $\downarrow 3.70\%$  | 37.80     | 33.40 | 35.60 | $\downarrow 10.37\%$ |
|                   |         |              | 8       | 75.97 | 33.07   | 54.52 | $\downarrow 4.12\%$  | 37.80     | 33.60 | 35.70 | $\downarrow 10.11\%$ |
|                   |         |              | 16      | 75.66 | 33.30   | 54.48 | $\downarrow 4.18\%$  | 37.80     | 33.60 | 35.70 | $\downarrow 10.11\%$ |
|                   |         |              | 32      | 75.59 | 33.30   | 54.45 | $\downarrow 4.25\%$  | 37.80     | 33.60 | 35.70 | $\downarrow 10.11\%$ |

# Online Calibration Hybrid Approach

- This hybrid approach combines between the greedy online approach and the uniform scheduling
- An anchor is always placed if the RQE exceeds the threshold value and there is still anchor budget left
- At each position for a uniform schedule  $i$ , it checks if the number of total anchors placed exceeds  $i$  or not. If it does not exceed it then it adds an anchor at the uniform position
- This approach ensures that the budget is fully used across timesteps

# Hybrid Online Calibration Results

Table 6: LLaDA-8B-Instruct accuracy under online-calibrated hybrid anchor scheduling.

| Model             | Setting | Method       | Anchors | Math  |         |       |                      | Code      |       |       |                      |
|-------------------|---------|--------------|---------|-------|---------|-------|----------------------|-----------|-------|-------|----------------------|
|                   |         |              |         | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval | MBPP  | Avg   | $\Delta\%$           |
| LLaDA-8B-Instruct | FP16    | Base         | —       | 78.85 | 34.87   | 56.86 | —                    | 39.63     | 39.80 | 39.72 | —                    |
|                   |         | DeltaQuaRot  | 1       | 71.11 | 28.58   | 49.85 | $\downarrow 12.33\%$ | 31.10     | 36.40 | 33.75 | $\downarrow 15.03\%$ |
|                   |         |              | 2       | 70.28 | 29.08   | 49.68 | $\downarrow 12.63\%$ | 35.98     | 37.00 | 36.49 | $\downarrow 8.14\%$  |
|                   |         |              | 4       | 70.81 | 28.73   | 49.77 | $\downarrow 12.47\%$ | 34.76     | 37.40 | 36.08 | $\downarrow 9.17\%$  |
|                   |         |              | 8       | 69.90 | 29.66   | 49.78 | $\downarrow 12.45\%$ | 37.20     | 37.60 | 37.40 | $\downarrow 5.85\%$  |
|                   |         |              | 16      | 71.04 | 29.22   | 50.13 | $\downarrow 11.84\%$ | 37.20     | 37.60 | 37.40 | $\downarrow 5.85\%$  |
|                   |         |              | 32      | 71.42 | 28.58   | 50.00 | $\downarrow 12.06\%$ | 38.41     | 35.80 | 37.11 | $\downarrow 6.58\%$  |
|                   | W4A4    | DeltaDuQuant | 1       | 75.82 | 34.09   | 54.95 | $\downarrow 3.36\%$  | 39.02     | 32.60 | 35.81 | $\downarrow 9.84\%$  |
|                   |         |              | 2       | 77.10 | 33.04   | 55.07 | $\downarrow 3.14\%$  | 37.20     | 32.80 | 35.00 | $\downarrow 11.89\%$ |
|                   |         |              | 4       | 76.65 | 32.81   | 54.73 | $\downarrow 3.75\%$  | 37.80     | 33.40 | 35.60 | $\downarrow 10.37\%$ |
|                   |         |              | 8       | 75.66 | 33.22   | 54.44 | $\downarrow 4.26\%$  | 37.80     | 33.40 | 35.60 | $\downarrow 10.37\%$ |
|                   |         |              | 16      | 75.74 | 33.80   | 54.77 | $\downarrow 3.68\%$  | 37.80     | 33.00 | 35.40 | $\downarrow 10.87\%$ |
|                   |         |              | 32      | 75.59 | 33.33   | 54.46 | $\downarrow 4.22\%$  | 40.24     | 33.20 | 36.72 | $\downarrow 9.55\%$  |

# Using KL-Divergence Metric

- Start by loading two models: an unquantized FP16 model and the quantized model
- Compute the KL divergence (quantized logits||FP16 logits) over the masked positions
- After each timestep feed the input of the FP16 model to the quantized model
- Calculate the cumulative KL divergence curve across timesteps
- Place anchors at timesteps that split the KL divergence curve into equal cost segments

# KL-Divergence Cross Task Results

Table 8: LLaDA-8B-Instruct accuracy under cross-task KL-calibrated anchor scheduling.

| Model             | Setting      | Method      | Anchors | Math  |         |       |                      | Code                |       |                     |                      |
|-------------------|--------------|-------------|---------|-------|---------|-------|----------------------|---------------------|-------|---------------------|----------------------|
|                   |              |             |         | GSM8K | MATH500 | Avg   | $\Delta\%$           | HumanEval           | MBPP  | Avg                 | $\Delta\%$           |
| LLaDA-8B-Instruct | FP16         | Base        | –       | 78.85 | 34.87   | 56.86 | –                    | 39.63               | 39.80 | 39.72               | –                    |
|                   |              |             | 1       | 71.11 | 28.58   | 49.85 | $\downarrow 12.33\%$ | 31.10               | 36.40 | 33.75               | $\downarrow 15.03\%$ |
|                   |              | DeltaQuaRot | 2       | 70.74 | –       | –     | –                    | 31.10               | 36.80 | 33.95               | $\downarrow 14.53\%$ |
|                   |              |             | 4       | 71.42 | –       | –     | –                    | 31.10               | 36.60 | 33.85               | $\downarrow 14.78\%$ |
|                   |              |             | 8       | 71.11 | –       | –     | –                    | 31.10               | 36.40 | 33.75               | $\downarrow 15.03\%$ |
|                   |              |             | 16      | 70.13 | –       | –     | –                    | 32.32               | 36.20 | 34.26               | $\downarrow 13.75\%$ |
|                   |              |             | 32      | 68.92 | –       | –     | –                    | 33.54               | 36.20 | 34.87               | $\downarrow 12.22\%$ |
|                   |              |             | W4A4    | 1     | 75.82   | 34.09 | 54.95                | $\downarrow 3.36\%$ | 39.02 | 32.60               | 35.81                |
|                   | 2            | 75.89       |         | –     | –       | –     | 39.02                | 32.60               | 35.81 | $\downarrow 9.84\%$ |                      |
|                   | DeltaDuQuant | 4           |         | 75.44 | –       | –     | –                    | 39.02               | 32.60 | 35.81               | $\downarrow 9.84\%$  |
|                   |              | 8           |         | 75.97 | –       | –     | –                    | 39.63               | 32.60 | 36.12               | $\downarrow 9.07\%$  |
|                   |              | 16          |         | 75.13 | –       | –     | –                    | 39.02               | 32.80 | 35.91               | $\downarrow 9.59\%$  |
|                   |              | 32          |         | 74.91 | –       | –     | –                    | 39.02               | 33.00 | 36.01               | $\downarrow 9.33\%$  |

# Summary of Best Results

LLaDA-8B-Instruct summary: FP16 base, plain rotation quantization, single-anchor delta-quant, and the best-improving reanchoring configuration found for each method

| Model             | Method       | Config                       | Math  |         |       |            | Code      |       |       |            | Overall |            |
|-------------------|--------------|------------------------------|-------|---------|-------|------------|-----------|-------|-------|------------|---------|------------|
|                   |              |                              | GSM8K | MATH500 | Avg   | $\Delta\%$ | HumanEval | MBPP  | Avg   | $\Delta\%$ | Acc     | $\Delta\%$ |
| LLaDA-8B-Instruct | FP16         | Base                         | 78.85 | 34.87   | 56.86 | –          | 39.63     | 39.80 | 39.72 | –          | 48.29   | –          |
|                   | QuaRot       | -                            | 67.55 | 25.34   | 46.45 | ↓18.32%    | 26.83     | 32.00 | 29.41 | ↓25.93%    | 37.93   | ↓21.45%    |
|                   | DeltaQuaRot  | 1 anchor                     | 71.11 | 28.58   | 49.85 | ↓12.33%    | 31.10     | 36.40 | 33.75 | ↓15.03%    | 41.80   | ↓13.44%    |
|                   | DeltaQuaRot  | Online-hybrid, 16 anchors    | 71.04 | 29.22   | 50.13 | ↓11.84%    | 37.20     | 37.60 | 37.40 | ↓5.85%     | 43.77   | ↓9.37%     |
|                   | DuQuant      | -                            | 75.74 | 32.75   | 54.25 | ↓4.60%     | 35.37     | 29.80 | 32.58 | ↓17.96%    | 43.41   | ↓10.10%    |
|                   | DeltaDuQuant | 1 anchor                     | 75.82 | 34.09   | 54.95 | ↓3.36%     | 39.02     | 32.60 | 35.81 | ↓9.84%     | 45.38   | ↓6.03%     |
|                   | DeltaDuQuant | Offline-WikiText2, 4 anchors | 76.95 | 33.59   | 55.27 | ↓2.79%     | 41.46     | 33.00 | 37.23 | ↓6.26%     | 46.25   | ↓4.22%     |



# Plan for Next Week

- Analyze the activation distributions using the relative MAD and relative max absolute difference for the Dream model
- Run the delta quantization experiments on Dream Model
- Run clipping experiments on Llada model